

PBSMS

Frontend Design Specification

Version 1.2 · Companion to SRS v2.1, Chapter 34

Complete interface proposal across all system modules

Item	Value
Document	PBSMS Frontend Design Specification
Version	1.2 — icon system specified (§4.5); v1.1's own token layer audited and completed
Status	For review. No part of this has been user-tested; see §15.
Governs	SRS v2.1 Chapter 34 (Frontend Architecture, UX & Offline/PWA Strategy) and Chapter 42 (Accessibility)
Scope	All modules across SRS Volumes 0–V, expressed as four frontend surfaces
Companion prototype	pbsms-frontend-prototype.html, manifest.json, sw.js — interactive, PWA-capable when served (not opened as a local file — see §9.2). Placeholder Unicode glyphs replaced with the §4.5.4 icon register as of this revision.

Revision Note — v1.1 → v1.2

v1.1 closed seven gaps found by auditing this document against the SRS. This revision closes one found by auditing v1.1 against itself. Chapter 34.2 requires central design tokens; v1.1 supplied colour, typography, space and breakpoints, then named IconButton in §5 without ever stating where an icon comes from, what size it is, what licence it carries, or what any given glyph means. Every component in §5 depends on that answer, and §11.1's commitment that colour is never the only carrier of meaning cannot be checked without it. As with v1.1, nothing here is a new feature — it is an existing requirement that was only partly satisfied.

#	Requirement	Was	Now
1	34.2 — central design tokens	Colour, type, space and breakpoints defined with real values; icons left undefined entirely	Icon source and licence, size and stroke tokens, and a currentColor rule (§4.5.1–4.5.3)
2	WCAG 1.4.1 / §11.1 — colour never the only carrier of meaning	Committed in prose, with no glyph-to-meaning mapping to check the commitment against	A fixed semantic register: glyph, colour token, and a mandatory text pairing per state (§4.5.4)
3	§5 — component library	IconButton and the Domain group referenced icons with no stated source	Both defer to §4.5; a component does not choose its own icon source, size or colour (§5)
4	§12 — performance budget	Icon payload uncounted, on surfaces budgeted to 150KB total	Counted inside the JS figure, with per-surface distinct-glyph caps (§12)

#	Requirement	Was	Now
5	§13 — build order	Icons unsequenced; would have arrived after the components that consume them	Stage 1, with tokens, for the reason stage 1 exists (§13)
6	Licence provenance	Unstated. Nothing prevented an incompatible asset entering the repo	ICONS.md plus an icons:audit CI gate that fails the build (§4.5.6)

No new deviations from the SRS arise from this revision. §14 is unchanged.

Revision Note — v1.0 → v1.1 (retained)

v1.0 was audited against SRS v2.1 Chapter 34 and Chapter 42 line by line. Six requirements were not met as written, and one design decision deviated from the SRS's stated pattern without being flagged as a deviation. That revision closed all seven. Nothing in this list was a new feature — every item was already a stated requirement in v1.0's own source document; v1.0 simply didn't satisfy it yet.

#	Requirement	Was	Now
1	34.1 — active campus visible	Context switcher showed School only	Switcher shows School and Campus (§6.1)
2	34.1 — breadcrumbs, responsive drawer	Named in the component library, never specified	Breadcrumb pattern and a dedicated Staff Console small-screen spec (§6.1.1)
3	FR-UX-020 — offline score entry	Only attendance was detailed; scores were a one-line inventory entry	Full detailed spec, same rigor as the register (§8.11)
4	FR-UX-010 / FR-UX-040 — PWA, data-saver	Asserted in prose, not designed; prototype had no manifest or service worker	Manifest, service-worker caching strategy, and a concrete data-saver mode specified (§9.2, §9.3); companion manifest.json and sw.js shipped with the prototype
5	34.5 — six required UI states	Three of six (loading, empty, restricted) had no spec or example	All six specified with examples (§8.12)
6	NFR-ACC-020 — CI-enforced accessibility gate	Written in this document; absent from the actual CI pipeline	Added to .github/workflows/ci.yml in the code scaffold (§11.2)
7	34.5 — hide vs. disable unavailable actions	Platform Console disabled blocked actions instead of hiding them, presented as compliant	Named explicitly as an approved, reasoned deviation (§14) rather than left unflagged

1. Purpose & Relationship to the SRS

SRS v2.1 Chapter 34 specifies frontend behaviour — an application shell, a design-token layer, offline capability for attendance and score entry, WCAG 2.1 AA, and distinct UI states. It does not specify a single screen. This document fills that gap and nothing else: it proposes the interface, and it defers to the SRS on every functional question.

Where this document and the SRS disagree, the SRS wins. Where this document proposes something the SRS does not cover, it is flagged as a proposal, not a requirement.

1.1 What this document contains

- A design-token system with real values (§4) — Chapter 34.2 requires tokens but defines none.

- An icon system: source, licence, size and stroke tokens, and a fixed glyph-to-meaning register (\$4.5).
- Four frontend surfaces (\$3), because 'the frontend' is not one product.
- A complete screen inventory across every module in SRS Volumes 0–V (\$7).
- Detailed specifications for the highest-risk screens (\$8).
- Offline and sync UX, satisfying FR-UX-010..040, with PWA infrastructure and data-saver mode (\$9).
- Role-based visibility rules mapped to Chapter 13 scope (\$10).
- Accessibility implementation detail against WCAG 2.1 AA (\$11).
- A performance budget tied to NFR-PERF-020 (\$12).
- Every deviation from the SRS's literal text, named and justified rather than left implicit (\$14).

2. Design Principles

These are derived from constraints in the SRS and in the market, not from general design preference. Each one is falsifiable — you can tell whether a screen obeys it.

Principle	Derived from	What it forbids
The register is the product	Attendance and score entry are the highest-frequency actions in the system (FR-ATT-010, FR-ASM-020)	Burying score entry three levels deep behind an academics menu. A teacher reaches either in one tap from their home screen.
Never lie about sync	FR-UX-030	Optimistic checkmarks. A row is 'saved on this phone' or 'saved to the school' and the interface always says which.
Money is auditable on screen	Chapter 23.1 — every balance derived from valid transactions	A balance shown as a single number with no path to the transactions that produced it.
Published is different from draft	FR-RES-030 — published results are never overwritten	Any visual treatment where a draft result and a published result look alike.
Data before decoration	NFR-PERF-020 (2.5s on 3G), FR-UX-040 (low data)	Hero images, icon fonts, decorative illustration, webfont stacks over 100KB.
One tenant at a time	TEN-012, TEN-013	Any screen that mixes two tenants' data, including in platform support tooling.

3. The Four Surfaces

The single most consequential proposal in this document: PBSMS is not one interface. It is four, sharing one design system, one component library and one codebase. Designing them as one app produces a desktop admin tool that teachers cannot use on a phone in a classroom — which is the failure mode that kills school software adoption.

Surface	Primary user	Device reality	Design posture
Staff Console	Head, administrator, accountant, coordinator, exam officer, admissions officer	Laptop or shared desktop, office, reliable power	Data-dense. Tables, filters, bulk actions, keyboard shortcuts. Desktop-first, responsive down.
Teacher Field App	Class and subject teachers	Personal Android phone, in a classroom, connectivity unreliable, one hand free	Offline-first. One task per screen, large touch targets, thumb-zone actions, works fully offline.

Surface	Primary user	Device reality	Design posture
Parent View	Guardians	Low-cost Android, metered data, often arrives from a WhatsApp link	Read-mostly, extremely light. No app install required. Loads under 2.5s on 3G or it has failed.
Platform Console	PBSMS staff (Chapter 3 roles)	Laptop, internal	Deliberately austere. Tenant-scoped, audit-visible, no bulk destructive actions.

These share: tokens, components, auth, and the offline/sync layer. They do not share navigation, information density, or layout. A parent never sees a sidebar; a teacher never sees a data table.

4. Design Tokens

Chapter 34.2 requires 'central design tokens' governing colour, typography, spacing and breakpoints, with a per-tenant branding layer on top. Here they are. Every value below is a proposal open to change — but the system needs concrete values, and 'we will decide later' is how design debt starts.

4.1 Colour — rationale

The palette comes from the subject's own materials: the deep green-black of a classroom blackboard, the warm manila of an exercise book, and the gold that already signals attainment in Ghanaian schools and on the national flag. It is deliberately not a generic SaaS blue.

Gold is reserved. It marks attainment and nothing else — a passing grade, a completed term, a settled account. If gold starts appearing on buttons and headers it stops meaning anything, and the one visual signal a parent scanning a report card actually needs is lost.

Core

Token	Value	Use	Contrast on canvas
--pb-ink	#12211F	Body text, headings	14.8:1 — AAA
--pb-ink-muted	#5A6663	Secondary text, table meta, captions	5.4:1 — AA
--pb-teal-900	#0B4F4A	Primary actions, headers, active nav	9.1:1 — AAA
--pb-teal-600	#12726B	Hover, links, focus ring	5.9:1 — AA
--pb-teal-50	#E6EFEC	Selected rows, subtle fills	—
--pb-gold	#9A6700	Attainment only — grades, settled, completed	5.2:1 — AA
--pb-gold-50	#FBF3E0	Attainment fills, award chips	—
--pb-canvas	#FAF7F0	Page background (warm, paper)	—
--pb-surface	#FFFFFF	Cards, tables, sheets	—
--pb-rule	#E3DED2	Hairlines, table borders, dividers	—

Every contrast ratio in this table is calculated, not tool-verified. Re-check each one against a contrast checker before implementation — see §11.2 and §15.

Semantic

Token	Value	Meaning in this product
--pb-success	#1E6B3A	Synced, posted, approved, present

Token	Value	Meaning in this product
--pb-warning	#8A5A00	Pending review, overdue soon, unsynced
--pb-danger	#A32017	Failed, reversed, absent, overdue, validation error
--pb-info	#12726B	Neutral system state

Colour is never the only carrier of meaning (WCAG 1.4.1). Every state above pairs with a shape, an icon, or text — see §4.5.4 for the glyph mapping and §11.1 for the conformance commitment.

4.2 Typography

Two faces, one of them free. The body face is the device's own system stack, which costs zero bytes and renders familiarly on cheap Android — a direct concession to FR-UX-040. One display face is loaded, subset, for headings and figures.

Role	Face	Why
Display / headings	Archivo (variable, subset to Latin, woff2, ~28KB)	A grotesque with enough width and weight range to carry dense admin headers without a second face. Not a high-contrast serif — those read as editorial, and this is a working tool.
Body / UI	system-ui stack — Roboto on Android, SF on iOS, Segoe on Windows	Zero network cost, instantly familiar, no FOUT on the screens teachers use most.
Figures & identifiers	Archivo with font-variant-numeric: tabular-nums; identifiers in ui-monospace	Scores, money and attendance counts must align in columns. Admission numbers, invoice numbers and BECE index numbers are read aloud and cross-checked against paper — monospace makes transcription errors visible.

Type scale (1.200 minor third, 16px base)

Token	Size	Weight	Use
--pb-text-2xs	11px	500	Table meta, timestamps
--pb-text-xs	12px	500	Labels, chips, captions
--pb-text-sm	14px	400	Table cells, secondary body
--pb-text-base	16px	400	Body. Never smaller for parent-facing text.
--pb-text-lg	19px	500	Card titles, section heads
--pb-text-xl	23px	600	Page titles
--pb-text-2xl	28px	600	Dashboard figures
--pb-text-3xl	33px	700	Teacher app primary figures, parent balance

4.3 Space, radius, elevation, motion

Group	Tokens
Space (4px base)	--pb-space-1 4px · -2 8px · -3 12px · -4 16px · -5 24px · -6 32px · -7 48px · -8 64px
Radius	--pb-radius-sm 4px (inputs, chips) · -md 8px (cards, sheets) · -lg 12px (modals) · -full (avatars only)

Group	Tokens
Elevation	--pb-shadow-1 0 1px 2px rgba(18,33,31,.08) · -2 0 4px 12px rgba(18,33,31,.10) · -3 0 12px 32px rgba(18,33,31,.14). Three levels only; anything needing a fourth is a layout problem.
Motion	--pb-ease standard cubic-bezier(.2,0,0,1) · --pb-dur-fast 120ms (state) · -base 200ms (sheets, drawers) · -slow 320ms (page transitions). All motion suppressed under prefers-reduced-motion.
Breakpoints	--pb-bp-sm 480px · -md 768px · -lg 1024px · -xl 1280px. Teacher app and Parent view are designed at 360px first — the real floor for Android in this market.

4.4 Tenant branding layer

Per Volume 0 Chapter 6, branding is per tenant. Exactly three things are tenant-overridable, resolved once at the layout root and never per screen:

- --pb-brand-primary — replaces teal in nav, primary buttons and document letterheads. Validated on upload for 4.5:1 contrast against white; rejected with a clear message if it fails, because a tenant must not be able to configure their own school into an inaccessible interface.
- Logo — used in the shell header, report cards, receipts and certificates.
- School display name and letterhead block.

Everything else — type, spacing, semantic colour, component shape, and the icon set of \$4.5 — is fixed. Tenants get identity, not a theme engine. This keeps the component library testable and stops a school from breaking its own accessibility conformance.

4.5 Icon System — fixes a v1.1 gap

Chapter 34.2 requires central design tokens governing colour, typography, spacing and breakpoints. v1.1 delivered all four and stopped. Icons are the fifth token group in everything but name: they carry attendance state on the register (§8.1), sync state in the SyncLedger (§8.2), result state on the report card (§8.7), and the entire meaning layer that §11.1 promises will survive being read in greyscale. v1.1 named IconButton in §5 and never said where an icon comes from, how large it is, what colour it may take, what it is allowed to mean, or what licence it ships under. This section is the missing half of an existing requirement, not a new one.

4.5.1 Source of record

Decision	Value	Why
Base set	Lucide — MIT licence, version pinned in package.json	Roughly 1,600 glyphs on a single 24px authoring grid with one stroke convention. MIT clears the entire set in one review rather than per file, which is exactly what sourcing icons individually from an aggregator fails to do.
Delivery	Vendored into packages/ui/icons as tree-shaken React components	No CDN dependency and no runtime fetch, so an offline Teacher Field App has its icons and a Parent View route never downloads a glyph it does not draw.
Prohibited	Icon fonts; a sprite sheet fetched as a separate request on Parent View; runtime CDN loading; a second general-purpose icon set	§2 forbids icon fonts outright, and §12 caps Parent View at 150KB. Two general sets means two stroke conventions, and the register's five attendance glyphs immediately stop reading as one family.

A note on aggregators, since the question will recur. Sites such as SVG Repo host assets under mixed per-file terms — CCO, MIT, Apache-2.0, GPL, MPL, OFL, Creative Commons non-commercial variants, and per-company 'logo' terms that vary by copyright holder. The licence travels with the file, not with the site, and the site disclaims liability for misuse.

Non-commercial and logo-licensed assets cannot ship in a commercial multi-tenant product at all, and copyleft assets raise a question this codebase does not need to answer. Aggregators are therefore permitted only under §4.5.5, only for glyphs the base set does not carry, and only under the three licences named there.

4.5.2 Size and stroke tokens

Token	Size	Stroke	Use
--pb-icon-sm	16px	1.5px	Table cell affordances, chips, StatusPill, inline meta
--pb-icon-md	20px	1.5px	Default. Buttons, form fields, sidebar nav, breadcrumb separators
--pb-icon-lg	24px	2px	BottomNav (§6.2), Teacher Field App row controls, TopBar
--pb-icon-xl	32px	2px	EmptyState and Restricted marks (§8.12)

The stroke steps at 24px so optical weight stays constant across sizes. A 1.5px stroke at 32px reads as a hairline beside --pb-text-lg; a 2px stroke at 16px fills its own counters and becomes a blob on a 360px panel, which is the screen this product is most often read on. Every size sits on the 4px base of §4.3, so an icon never breaks a row's vertical rhythm.

Icons are never scaled by CSS transform, and an icon rendered at a size with no token is a review defect. There are four sizes because four is enough; a fifth is a layout problem, in the same spirit as the three-level elevation rule.

4.5.3 Colour

Every icon inherits currentColor. No fill or stroke colour is hardcoded in any SVG in the package. This is enforced by the gate in §4.5.6, not left to convention.

That single rule is what lets one set do three jobs without a second set:

- Semantic state comes from the parent element, so the tokens in §4.1 — --pb-success, --pb-warning, --pb-danger, --pb-info — reach icons with no icon-specific colour layer.
- Tenant branding (§4.4) reaches nav and primary-button icons for free. No per-tenant icon assets, no rebuild when a school changes its colour.
- Greyscale legibility becomes testable rather than asserted: strip colour and the glyph shape survives, and the shape is what carries the meaning.

Gold is reserved for icons on the same terms as everywhere else in §4.1 — attainment only. A gold badge-check on a settled invoice or a published result; never a gold check on a saved form.

4.5.4 Semantic register

One glyph per meaning, fixed across all four surfaces. A teacher learns the offline model once (§8.11); they should learn each glyph once as well.

Meaning	Glyph	Colour token	Text pairing — always present
Present	check	--pb-success	'Present' / 'P'
Absent	x	--pb-danger	'Absent' / 'A'
Late	clock	--pb-warning	'Late' / 'L'
Excused	file-text	--pb-ink-muted	'Excused' / 'E'
Sick	heart-pulse	--pb-ink-muted	'Sick' / 'S'
Saved on this phone	smartphone	--pb-warning	'on this phone'
Saved to school	cloud-check	--pb-success	'saved to school'

Meaning	Glyph	Colour token	Text pairing — always present
Sending	refresh-cw, rotating; static under prefers-reduced-motion	--pb-info	'Sending — n of m saved'
Could not be saved	alert-triangle	--pb-danger	'could not be saved' + reason
Locked after submission	lock	--pb-ink-muted	'Locked' + 'Request reopen'
Restricted	shield-off	--pb-ink-muted	Named restriction and who to ask (§8.12)
Draft result	file-pen	--pb-ink-muted	'Draft'
Published result	badge-check	--pb-gold	'Published' + date
Attainment / award	award	--pb-gold	The grade, in text

The right-hand column is the load-bearing one. §11.1 commits to colour never being the sole carrier of meaning; this table is where that commitment becomes checkable instead of aspirational. A glyph added to this table without a text pairing has not met WCAG 1.4.1, and adding one is a review failure rather than a style preference.

4.5.5 Domain glyphs

Four meanings the base set does not carry, and cannot be expected to:

Glyph	For	Note
Mobile money	MoMo as a payment method (§8.5, FR-PAY-010)	The method most guardians in this market will actually use. A generic wallet glyph reads as neither MoMo nor card, and the distinction is the first thing a cashier looks for.
Ghana cedi mark	CurrencyInput, BalanceSummary, receipts	GH¢ is set as text in Archivo tabular figures wherever it can be. A glyph exists only for the cases where it must sit inside a control.
Curriculum strand	NaCCA strand and indicator tree (DOM-010, §7.14)	Must distinguish strand from sub-strand from indicator at --pb-icon-sm, which is the size the tree actually renders at.
Index number	BECE candidate registration (DOM-040)	Should read as an identifier meant to be transcribed against paper, matching the ui-monospace treatment §4.2 gives those numbers.

Rules for anything entering this list, drawn or sourced:

- Same 24px authoring grid, same stroke widths, same terminal and join style as the base set. A domain glyph that looks bought in is a defect, not a variation.
- `currentColor` only. No hardcoded fill or stroke, no embedded raster, no gradient.
- Committed to packages/ui/icons/domain/ with a provenance row in ICONS.md (§4.5.6).
- Sourced glyphs are permitted only under CC0, MIT or Apache-2.0, verified per file at the original source rather than at the aggregator, and normalised to the grid before commit. CC BY, CC BY-NC, GPL, MPL, OFL and per-company logo terms are rejected outright — attribution-bearing licences included, because a licence that requires attribution in a product with no credits screen is a compliance liability rather than a free asset.

4.5.6 Provenance and enforcement

A licence rule written in prose is worth nothing at the moment someone is in a hurry and needs one more glyph. This one is a gate, in the same shape as the RLS coverage gate: a rule CI can fail a build on.

ICONS.md, committed beside the package, records the base set name, its pinned version, its SPDX identifier, and one row per domain glyph: file, meaning, origin (drawn, or the original source URL — not the aggregator's), SPDX identifier, date added, and who normalised it.

The icons:audit CI job fails the build when any of the following is true:

- an SVG under packages/ui/icons/ has no corresponding provenance row in ICONS.md;
- an SVG carries a literal fill or stroke colour instead of currentColor;
- an SVG's viewBox is not the 24px authoring grid;
- the installed base-set version has drifted from the version recorded in ICONS.md.

This runs alongside the pa11y-ci accessibility job added in §11.2 and the performance budget check in §12. All three are starting gates from the first frontend commit rather than additions once a component library exists, for exactly the reason §11.2 gives about retrofitting.

4.5.7 A note on the figures in this document

The ASCII figures in §6.2, §8.1, §8.2, §8.11 and §8.12 use Unicode stand-ins —  — because they have to render inside a text document. They are not the shipped glyph set and should not be read as design intent.

The companion prototype previously carried those same stand-ins in live HTML. As of this revision it renders the §4.5.4 register as inline SVG instead, which is also the cheapest available test of whether that register covers every state the daily register screen needs. It did not, on the first pass: 'excused' and 'sick' had no distinct glyph and were reaching the screen as bare letters with no icon at all — a 1.4.1 failure that was invisible for as long as the figure was ASCII.

5. Component Library

Built once, used across all four surfaces. Naming follows what users control, not how it is built (a person manages notifications, not webhook config).

Every icon rendered by any component in this library comes from the set specified in §4.5, at a size token from §4.5.2, coloured by currentColor from its parent. A component does not choose its own icon source, size or colour, and a component that needs a glyph the semantic register does not carry raises an amendment to §4.5.4 rather than importing one locally.

Group	Components
Foundational	Button (primary/secondary/ghost/danger), IconButton (§4.5), Link, Chip, Badge, Avatar, Tag, Divider, Spinner, ProgressBar
Form	TextField, NumberField (tabular figures), Select, Combobox, DatePicker (term-aware), Checkbox, Radio, Toggle, Textarea, FileUpload, ScoreInput (bounded, keyboard-fast), CurrencyInput (GHS-aware), SearchField
Data	DataTable (sortable, filterable, paginated, column-configurable), DefinitionList, KeyFigure, TrendSparkline, EmptyState, StatusPill, AuditTrail, Timeline, ComparisonBar
Navigation	AppShell, Sidebar, TopBar, ContextSwitcher (school/campus/year/term), Breadcrumbs, Tabs, Stepper, BottomNav (mobile), CommandPalette
Feedback	Toast, InlineAlert, ConfirmDialog, DestructiveConfirm (types-to-confirm), FieldError, FormSummaryError, OfflineBanner, SyncLedger
Surface	Card, Sheet (mobile bottom sheet), Modal, Drawer, Popover, Tooltip, Accordion
Domain	RegisterRow, ScoreGrid, ResultCard, GradeChip, InvoiceLine, PaymentAllocator, BalanceSummary, ReportCardPreview, CompetencyMeter, ApprovalBar, ConsentToggle

The Domain group is where this product differs from any generic admin template. A ScoreGrid that understands assessment weighting, a PaymentAllocator that splits one payment across invoice lines, and a CompetencyMeter that renders NaCCA indicator coverage cannot be bought off the shelf and should be built with the most care. The bespoke

glyphs these components depend on — mobile money, the cedi mark, the curriculum strand tree, the BECE index number — are specified in §4.5.5 and are the only drawing this document requires.

6. Navigation & Information Architecture

6.1 Staff Console

Sidebar grouped by how school staff actually think about their work, not by SRS volume order. Groups collapse; items are permission-generated (Chapter 34.1).

PBSMS [School ▼] [Campus ▼] [2026/2027 ▼] [Term 2 ▼] %K 🛎 Ama A. ▼

△ Today ← role-specific landing

PEOPLE

Admissions · Students · Guardians · Staff

LEARNING

Attendance · Assessment · Results · Report cards

Promotion · Curriculum (NaCCA) · BECE

MONEY

Fees · Invoices · Payments · Reconciliation

Assistance · Statements

COMMUNICATION

Messages · Templates · Delivery log · Consent

OPERATIONS

Library · Transport · Health · Discipline · Inventory

INSIGHT

Analytics · Group roll-up ← only if tenant has >1 school

⊙ Settings · Users & roles · Branding · Integrations · Subscription · Audit log · Data requests

The context switcher is four-part — school, campus, academic year, term — not three. Chapter 34.1 requires the active campus to be visible, not only the active school; a tenant with one school and one campus can collapse the first two into a single control, but the underlying state (which campus is active) always exists and is always shown once a tenant has more than one, because that is exactly the case a single collapsed control would hide.

Fixed v1.0 defect: the switcher previously showed School only. Campus is Chapter 2's actual isolation-adjacent unit below School, and Volume 0 exists specifically to support multi-campus tenants — a shell that cannot show which campus is active does not meet the requirement it claims to.

Breadcrumbs

Required by 34.1, absent from v1.0. Breadcrumbs sit directly under the top bar on any screen more than one level deep, and are themselves navigable — clicking 'JHS 2A' from a student profile returns to that class's roster, not to a generic student list.

Students > JHS 2A > Ama Mensah

On the Teacher Field App and Parent View, breadcrumbs do not appear — those surfaces are shallow by design (§3), and a back arrow in the header is sufficient. Breadcrumbs are a Staff Console and Platform Console pattern only.

6.1.1 Staff Console — small screens

34.1 requires 'a responsive drawer on small screens.' v1.0 named this in the surfaces table as 'desktop-first, responsive down' and never specified what 'down' meant. It means the following, concretely:

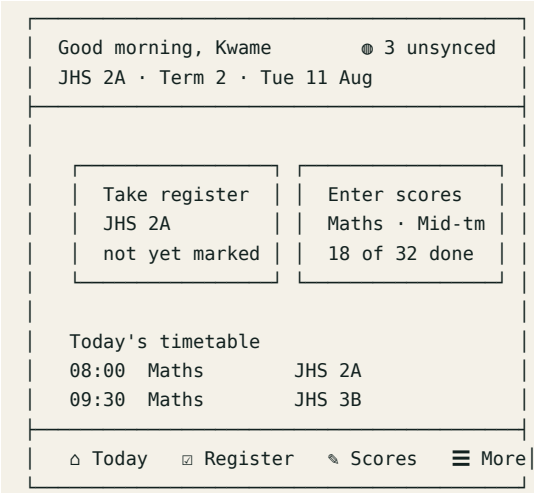
Breakpoint	Sidebar	Top bar	Tables
≥1024px (--pb-bp-lg)	Persistent, expanded	Full — all four context controls, search, notifications, user menu inline	Full DataTable, all columns
768-1023px (--pb-bp-md)	Collapses to icon rail; expands as an overlay on tap	Context switcher collapses into a single 'School · Campus · Term' summary button that opens a sheet	DataTable drops secondary columns behind a 'more' control per row
<768px (--pb-bp-sm)	Hidden by default; opens as a full-height Drawer (component, §5) triggered by a menu icon, closes on selection or scrim tap	Reduced to: menu icon, breadcrumb (current screen only), user avatar	Tables convert to stacked cards — one record per card, primary fields only, tap through for detail

The icon rail at the middle breakpoint is the one place in this product where an icon carries a label's whole job. Rail items therefore keep an accessible name and a tooltip on hover and focus, and the rail is not offered below 768px at all — at that width the Drawer restores the text labels rather than shrinking further.

This matters more than it looks: a proprietor checking collections from a phone between school visits is a real, common case for this product, not an edge case — Chapter 8's Proprietor role is explicitly the one most likely to be mobile.

6.2 Teacher Field App

No sidebar. Four bottom-nav destinations, sized for one-handed use, with the two highest-frequency tasks reachable in one tap from launch.



BottomNav glyphs render at --pb-icon-lg with a permanent text label beneath, never icon-only. The label is not optional at any width — a teacher using this app for the first time in a classroom cannot afford to guess, and the four destinations are fixed, so there is no width at which labels stop fitting.

6.3 Parent View

No navigation chrome at all on first load. A parent arriving from a WhatsApp link sees the thing they were notified about, immediately, with everything else one tap below it. Multiple children appear as a horizontal switcher, not a menu.

6.4 Platform Console

Tenant list → tenant detail. Every screen states which tenant is in view, in a persistent bar that is visually distinct from the tenant-facing product (dark, unmistakable) so PBSMS staff can never confuse the console with a customer's own account. Impersonation adds a second, louder banner naming the ticket and the expiry time (TEN-021).

7. Screen Inventory — All Modules

Complete coverage of SRS Volumes 0–V. Surface key: S = Staff Console, T = Teacher Field App, P = Parent View, X = Platform Console.

7.1 Platform & Tenancy (Volume 0)

Screen	Surface	SRS reference
Tenant list (search, status filter, plan, student count)	X	Ch 2, Ch 4
Tenant detail — profile, schools, subscription, health	X	Ch 2, Ch 5
Tenant lifecycle actions — suspend, offboard, reinstate	X	Ch 4.1
Plan & pricing configuration	X	Ch 5.1
Metering & usage — active students per tenant per period	X	TEN-030, TEN-031
Platform invoices & dunning status	X	FR-BIL-010..040
Impersonation request & active-grant banner	X	TEN-021
Platform audit log	X	TEN-022
Platform revenue dashboard (MRR, churn, ARPU)	X	FR-BIL-050

7.2 Onboarding & Tenant Settings

Screen	Surface	SRS reference
Signup / trial start	S	Ch 4.1
Onboarding wizard — school, year, terms, divisions, levels	S	FR-ONB-010
Starter data review (grading scale, fee categories, templates)	S	FR-ONB-010
Bulk import — upload, map columns, dry-run validation, commit	S	FR-ONB-020
Import error report & re-upload	S	FR-ONB-020
Branding — logo, primary colour, letterhead	S	Ch 6, Ch 34.4
Users & roles, delegation, conflict-of-interest warnings	S	Ch 13
Integrations — payment, messaging, credential health	S	Ch 36
Subscription & billing (tenant side)	S	TEN-031

7.3 Dashboards (Chapter 14)

Screen	Surface	SRS reference
Proprietor dashboard — group roll-up across schools	S	FR-ANL-010
Headmaster dashboard — approvals, attendance, collections	S	Ch 14.1

Screen	Surface	SRS reference
Accountant dashboard — today's collections, unverified, overdue	S	FR-FIN-040
Teacher home — register + scores + timetable	T	Ch 14.1
Parent home — child summary, balance, latest results	P	FR-STU-060
Student home — own results, timetable	P	FR-RES-040
KPI configuration	S	Ch 14.2
Action tracker & escalations	S	FR-COM-060

7.4 Admissions (Chapter 15)

Screen	Surface	SRS reference
Application list — state filter, ageing, source	S	FR-ADM-010
Application detail — identity, guardians, documents, medical	S	FR-ADM-010
Duplicate-match review (side-by-side, merge or dismiss)	S	FR-ADM-020
Interview / entrance assessment scheduling & outcome	S	FR-ADM-010
Approval & conversion to student (atomic, confirm step)	S	FR-ADM-030
Admissions KPIs — volume, processing time, conversion	S	FR-ADM-050

7.5 Students & Guardians (Chapter 16)

Screen	Surface	SRS reference
Student list — filter by class, status, division, campus	S	FR-STU-010
Student profile shell with tabs (below)	S	FR-STU-010
• Identity & enrolment history	S	FR-STU-010
• Guardians & relationship flags	S	FR-STU-020
• Academics — results by term and year	S	FR-RES-050
• Attendance — trend and exceptions	S	FR-ATT-050
• Finance — invoices, payments, balance	S	Ch 24
• Health (restricted access)	S	FR-OPS-030
• Discipline (restricted access)	S	FR-OPS-040
• Documents	S	FR-DOC-010
• Timeline — every significant event, chronological	S	FR-STU-050
Transfer — class, campus, inbound, outbound	S	FR-STU-030
Guardian directory & linked children	S	FR-STU-020

7.6 Academic Structure (Chapter 17)

Screen	Surface	SRS reference
Academic years & terms — states, windows	S	FR-ACA-020

Screen	Surface	SRS reference
Divisions, class levels, classes	S	FR-ACA-010
Subjects by division/level (+ optional NaCCA mapping)	S	FR-ACA-030, DOM-010
Teacher assignment & workload view	S	FR-ACA-050
Timetable builder with live conflict detection	S	FR-ACA-040
Timetable views — class, teacher, room	S	FR-ACA-040

7.7 Attendance (Chapter 18)

Screen	Surface	SRS reference
Daily register — offline-capable	T	FR-ATT-010
Register (desktop fallback)	S	FR-ATT-020
Correction with reason (retains original)	S	FR-ATT-030
Class & school attendance summary	S	FR-ATT-050
Absence follow-up queue & guardian contact	S	FR-ATT-040
Staff attendance	S	Ch 18.1
Child attendance view	P	FR-STU-060

7.8 Assessment, Grading & Results (Chapters 19–21)

Screen	Surface	SRS reference
Assessment configuration & weighting (must total 100%)	S	FR-ASM-010
Score entry grid — desktop, keyboard-fast	S	FR-ASM-020
Score entry — offline-capable mobile	T	FR-ASM-020, FR-UX-020
Grading scale & policy versions	S	FR-GRA-030, FR-GRA-070
Grading run — preview, blocking errors, no silent defaults	S	FR-GRA-060
Results workflow queue (draft → published)	S	FR-RES-010
Result review & return-with-comment	S	FR-RES-010
Publication gate — checklist of unmet conditions	S	FR-RES-020
Published results & version history	S	FR-RES-030
Reopen with reason & authority	S	FR-RES-030
Result analytics — trend, class, subject, division	S	FR-RES-050
Child results view	P	FR-RES-040

7.9 Documents & Promotion (Chapter 22)

Screen	Surface	SRS reference
Report card batch generation with progress	S	FR-DOC-010, NFR-PERF-023
Report card template editor & preview	S	FR-DOC-030

Screen	Surface	SRS reference
Certificate issue (JHS completion)	S	FR-PRO-010
Public document verification lookup (QR target)	P	FR-DOC-020
Promotion run — system recommendation vs decision	S	FR-PRO-010, FR-PRO-020
Promotion approval & next-year enrolment creation	S	FR-PRO-030

7.10 Finance (Chapters 23–25)

Screen	Surface	SRS reference
Fee structure builder — year, term, level, category	S	FR-FEE-010
Instalment plan & proration rules	S	FR-FEE-020, FR-FEE-030
Penalty rules	S	FR-FEE-040
Invoice generation run & preview	S	FR-PAY-040
Invoice list & detail with line items	S	Ch 24.1
Payment capture — cash, MoMo, card, transfer	S	FR-PAY-010
Payment allocation across invoice lines	S	Ch 24.1
Receipt issue & reprint	S	Ch 24.4
Financial assistance — scholarship, discount, waiver	S	FR-FIN-010
Reversal request & approval (maker-checker)	S	FR-FIN-020
Reconciliation workspace — provider vs internal	S	FR-FIN-030
Student statement of account	S	Ch 24.1
Finance dashboard	S	FR-FIN-040
Invoice, balance & pay now	P	FR-PAY-010

7.11 Communication (Chapter 26)

Screen	Surface	SRS reference
Compose — channel picker with cost estimate	S	FR-COM-040
Template library & version editor	S	FR-COM-020
Audience builder — class, level, arrears, custom	S	FR-COM-040
Send review — recipient count, channel split, cost	S	FR-COM-030
Delivery log — per recipient, per channel, retry state	S	FR-COM-040
Consent registry & withdrawal handling	S	DP-070
Message thread	P	FR-COM-060

7.12 Analytics (Chapter 27)

Screen	Surface	SRS reference
School analytics — attendance, performance, collections	S	FR-ANL-020

Screen	Surface	SRS reference
Group roll-up comparison across schools	S	FR-ANL-030
Cohort & trend explorer	S	FR-ANL-020
AI-drafted summary with human-approval gate	S	FR-ANL-040

7.13 Supporting Operations (Chapter 28)

Screen	Surface	SRS reference
Library — catalogue, circulation, fines	S	FR-OPS-010
Transport — routes, stops, vehicles, assignments	S	FR-OPS-020
Health — records, incidents, medication log	S	FR-OPS-030
Discipline — case, investigation, response, appeal	S	FR-OPS-040
Inventory — assets, stock, issuance, low-stock alerts	S	FR-OPS-050

7.14 Compliance & Curriculum (Volume V)

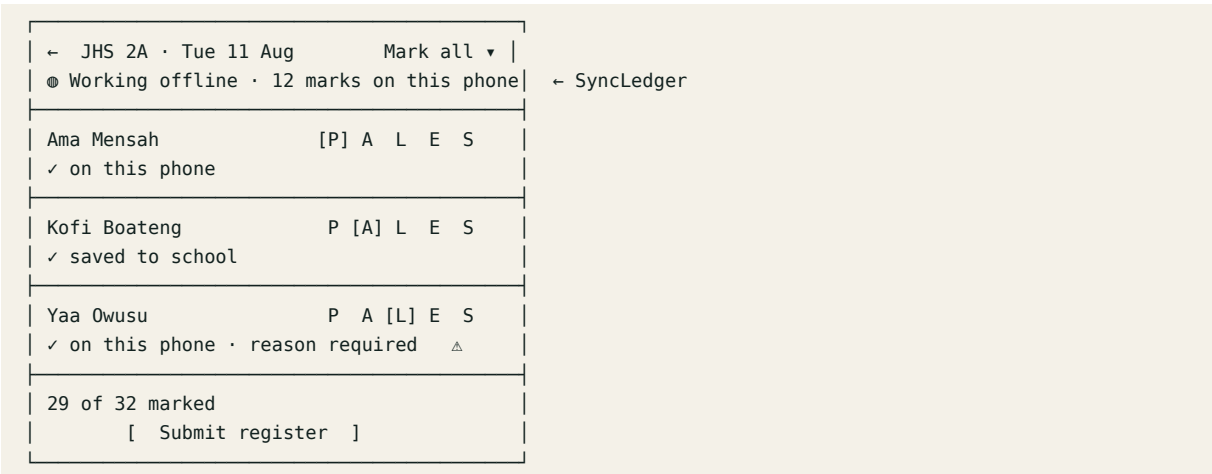
Screen	Surface	SRS reference
Consent registry per guardian per channel	S	DP-070
Data subject request queue with 30-day SLA timer	S	DP-030, DP-090
Retention & archive status by record type	S	Ch 40.2
Audit log viewer (tenant-visible, incl. platform actions)	S	TEN-022
NaCCA curriculum mapping — strand/indicator tree	S	DOM-010
Competency coverage report	S	DOM-020
BECE candidate registration & index numbers	S	DOM-040
Mock BECE results & readiness analytics	S	DOM-050, DOM-060
GES statutory return export	S	DOM-070
CSSPS choices & placement outcome	S	DOM-080

8. Key Screen Specifications

Ten screens carry most of the product risk — they are the highest-frequency, the highest-consequence, or the hardest to retrofit. Specified here in detail; the rest follow the patterns they establish.

8.1 Daily Register (Teacher Field App) — the signature screen

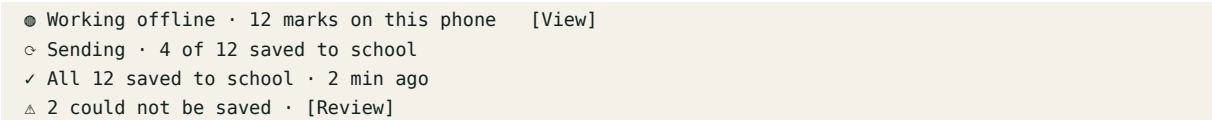
Highest-frequency action in the entire system, performed in a classroom, often with no connectivity. Everything about this screen is subordinate to one goal: a teacher marks 32 children in under 45 seconds and knows exactly what has and has not reached the school.



- Status is a five-state segmented control per row (Present, Absent, Late, Excused, Sick), not a dropdown — one tap, no modal, per FR-ATT-020.
- Each row states its own sync state in words, never a bare icon: 'on this phone' vs 'saved to school'. This is FR-UX-030 taken literally.
- The five status glyphs and the two sync glyphs come from the register in §4.5.4 and appear at --pb-icon-lg beside their letter, never instead of it. This row is the densest use of icons in the product and the one most likely to be read in bright sunlight on a cheap screen.
- The full roster is pre-fetched when the app last had connectivity, so the screen opens with data even in airplane mode.
- 'Submit register' works offline — it queues. The button never lies: offline it reads 'Submit when back online', and the SyncLedger count increments.
- Rows are 56px minimum, targets 44×44 minimum (WCAG 2.5.5), ordered by surname, sticky search for large classes.

8.2 SyncLedger (component, all Teacher Field App screens)

The proposed signature element of the product. Instead of a spinner, a persistent honest ledger of what has left the device.



- Always visible on register and score screens. Never a toast, because a toast disappears and the question 'did my marks save?' does not.
- Tapping it opens the queue: what is pending, what failed, what the failure was, and what to do about it.
- Conflicts (FR-ATT-011) surface here for manual reconciliation when two different users marked the same student — never silently resolved.
- Each of its four states pairs a distinct glyph from §4.5.4 with a text label and a colour token, so the state is legible without colour vision. The rotating 'sending' glyph is static under prefers-reduced-motion (§4.3) and its progress is carried by the text, not the rotation.

8.3 Score Entry Grid (Staff Console)

A spreadsheet-grade editing experience, because that is what it replaces and users will compare it to Excel.

- Keyboard-first: Enter moves down, Tab moves right, values commit on blur, no save button per cell.

- Bounds enforced at the input (FR-ASM-030); an out-of-range value is rejected inline with the valid range stated, not silently clamped.
- Missing scores require an explicit status — absent, exempt, pending — never left ambiguous.
- Weighting bar shows live total; publication is blocked, with the reason stated, while weights do not sum correctly (FR-ASM-010).
- Locked after submission with a visible lock state and a 'request reopen' action rather than a disabled control with no explanation (FR-ASM-040).

8.4 Results Publication Gate (Staff Console)

FR-RES-020 lists five conditions for publication. The screen renders them as a live checklist, not an error after the fact.

Publish · JHS 2A · Term 2 · 2026/2027

✓ All 9 subjects approved

✓ Class teacher review complete

✓ Headmaster approval recorded

△ 2 students missing Social Studies scores

Ama Mensah · Kofi Boateng [Resolve]

✓ Report template available

[Publish] – disabled, with the reason stated

A disabled button with no explanation is the most common failure in approval UI. Here the reason is always adjacent to the control, and every unmet condition links to the screen that resolves it.

8.5 Payment Allocation (Staff Console)

Chapter 24.1's model — payments independent of invoices, joined by allocations — is unusual and must be visible in the interface, or cashiers will misunderstand it.

- Left: the payment (amount, method, reference, provider status). Right: outstanding invoice lines.
- Method is shown with its own glyph — cash, MoMo (\$4.5.5), card, transfer — because the method drives what the cashier does next and is the field most often scanned rather than read.
- Allocation is explicit — drag or enter amounts per line, with a running unallocated remainder that must reach zero or be deliberately left as credit.
- Unallocated credit is a first-class visible state, not an error.
- Reversal is never a delete. The action reads 'Reverse payment', requires a reason, and produces a linked correcting entry shown in the same timeline (FR-FIN-020).

8.6 Parent Home (Parent View)

Assume arrival from a WhatsApp link on a metered connection. Under 2.5s on 3G, under 150KB total, no login wall for published content reached via an authenticated link.

Sunrise Basic School

● Ama ○ Kwesi

← child switch

Term 2 results are ready

[View report card]

Balance

GH¢ 450.00 due 30 Aug

[Pay now]

[See statement]

Attendance this term	96%
Present 44 · Absent 2 · Late 1	

Note what is absent: no sidebar, no dashboard grid, no KPI cards. A guardian has three questions — how is my child doing, what do I owe, and is anything wrong — and the screen answers them in that order. Parent View draws from a deliberately small subset of the icon register; §12 caps it, because every glyph is bytes on a metered connection.

8.7 Report Card (Parent View & print)

- One layout that renders on a 360px screen and prints to A4 without a separate template.
- Grade chips use gold for attainment, but the grade is always also stated in text.
- Where the tenant uses NaCCA (DOM-030), the competency profile appears above raw scores, because that is what the curriculum is actually assessing.
- The published version and date are always shown, marked with the published glyph in gold per §4.5.3. If a result was later revised, both versions are reachable — FR-RES-030 makes this a data guarantee and the UI must not hide it.

8.8 Reconciliation Workspace (Staff Console)

- Two columns: provider settlement vs internal records, with matched, unmatched and disputed lanes.
- Provider fees are shown as their own line (FR-FIN-030) — a settlement that is short by the fee is not a discrepancy and must not look like one.
- Every unmatched item has one obvious next action, never just a red flag.

8.9 Group Roll-Up (Staff Console, multi-school tenants only)

The capability the old single-school architecture could not express. Appears only when the tenant has more than one school (FR-ANL-030).

- Schools as rows, metrics as columns — collections, attendance, pass rate, outstanding actions.
- Ranked comparison with the tenant's own average as the baseline, not an industry benchmark that does not exist.
- Drill from any cell into that school's own dashboard, with the context switcher following the drill.

8.10 Impersonation Banner (Platform Console)

TEN-021 and TEN-022 require impersonation to be time-boxed, ticket-linked and visible to the tenant in real time. The interface must make it impossible to forget you are inside a customer's account.

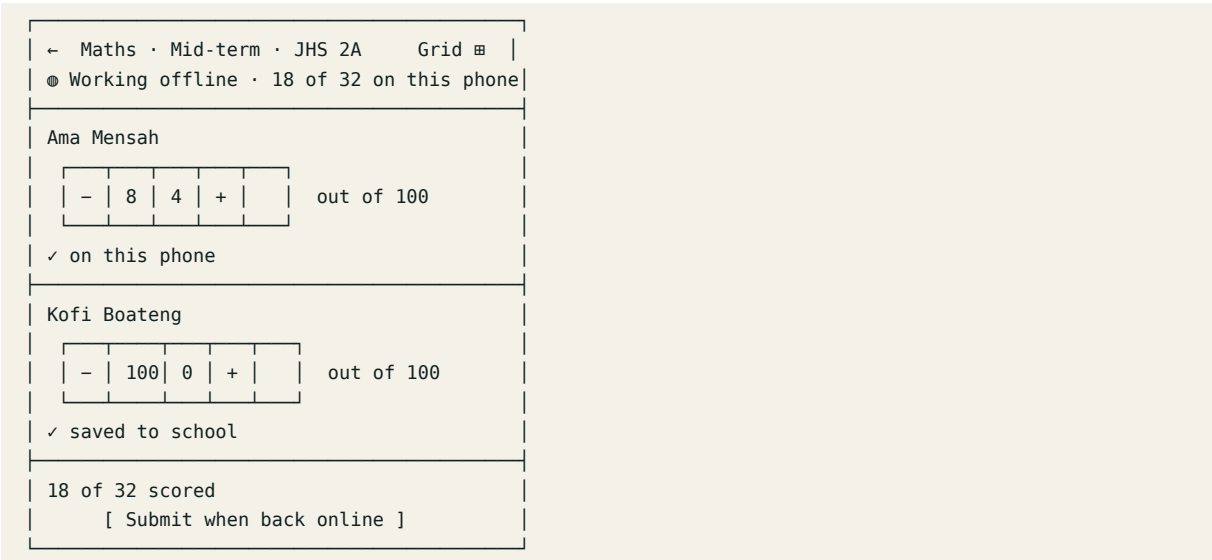
■ SUPPORT SESSION · Sunrise Basic School ■
Ticket #4821 · expires in 47 min · read-only
This session is visible in the school's audit log.
[End session now]

Full-width, high-contrast, non-dismissible, present on every screen for the duration. Actions blocked for the platform role (reversal, export, permission change) are visibly disabled with the four-eyes requirement named — a deliberate departure from 34.5's default pattern of hiding unavailable actions; see §14 for why, rather than leaving it unstated here.

8.11 Score Entry (Teacher Field App, offline)

FR-UX-020 names attendance and score entry together, with equal force: 'the two highest-frequency, most connectivity-sensitive workflows.' v1.0 gave attendance a full spec and the score-entry equivalent a single line in the inventory table.

That asymmetry was corrected in v1.1 — this screen reuses every mechanism the register established (pre-fetch, IndexedDB queue, SyncLedger) rather than inventing new ones, because a teacher should not have to learn two different offline behaviours in one app.

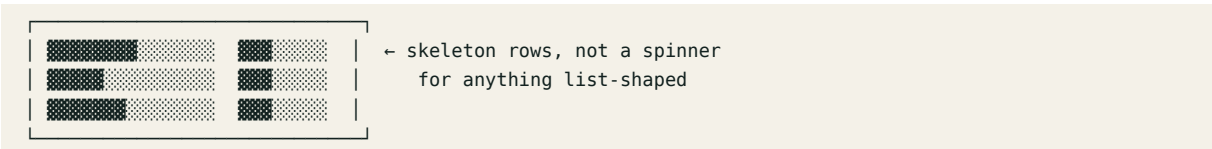


- One student per screen section, not a spreadsheet — the Staff Console grid (§8.3) is correct for a desktop mouse and keyboard; a numeric stepper with a large tap target is correct for a thumb. Same underlying data, two different input shapes for two different devices, per §3's design posture.
- The -/+ stepper is the primary input; a tap on the number opens the OS numeric keypad for direct entry, satisfying FR-ASM-030's bounds at input time on both paths.
- A 'Grid' toggle in the header switches to a compact all-students-one-screen view for a teacher who prefers scrolling over stepping, once scores have been reviewed once in the detailed view — an efficiency option, not the default.
- Reuses SyncLedger and the identical 'on this phone' / 'saved to school' language and glyphs from the register (§8.1–8.2, §4.5.4), so a teacher learns the offline model once.
- Bounds, missing-score status and the submitted lock (FR-ASM-030, FR-ASM-040) apply identically to this surface and to the Staff Console grid — one validation rule, enforced server-side, expressed in two UIs.

8.12 Loading, Empty & Restricted States

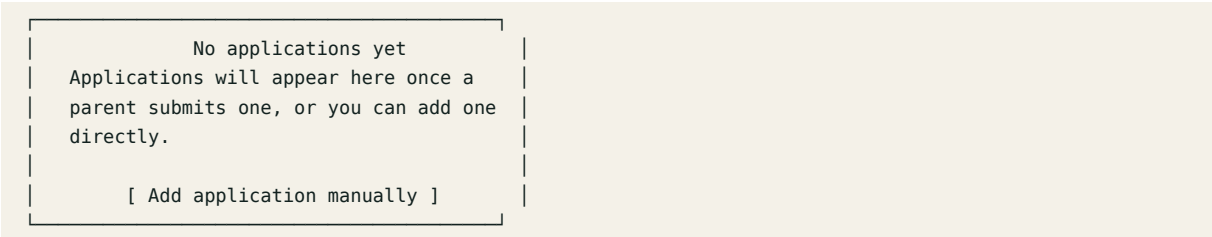
34.5 names six required states: loading, success, error, offline, empty, restricted. v1.0 specified and demonstrated success (§8.4, pills), error (§11.1, form errors) and offline (§8.1–8.2) in detail, but named the other three only as component names in the library (§5) with no actual specification. All six are specified here.

Loading



Skeleton screens matching the target layout's shape for lists and tables (DataTable, dashboards); a centred Spinner only for actions with no predictable shape (e.g. report-card batch generation, which already has its own named progress indicator per NFR-PERF-023). A skeleton must never persist past 5 seconds without becoming an error state — a frozen skeleton is a worse lie than a spinner.

Empty



Per the writing guidance this document follows: an empty screen is an invitation to act, not an apology. Every EmptyState names what will make the list stop being empty and, where a direct action exists (add a student, create an invoice), offers it in place. A results list empty because nothing has been marked yet reads differently ('Scores haven't been entered for this term') from one empty because a filter excludes everything ('No students match these filters — clear filters'). Where an EmptyState carries a mark, it is a single glyph at --pb-icon-xl in --pb-ink-muted, never an illustration (§2).

Restricted



Reached when a permission-generated menu correctly hid a link, but a direct URL or a stale bookmark reaches the screen anyway — the case 34.5's own principle predicts, since the server, not the menu, is the actual authority. States what is restricted and who to ask, never a bare 403 or a silent redirect to the dashboard, which reads as a bug rather than a boundary.

9. Offline & Sync UX

9.1 Attendance & Score Sync

Implements FR-UX-020 and FR-UX-030. Scope is deliberate: attendance and score entry are offline-capable. Nothing else is, in v1.

Concern	Behaviour
Pre-fetch	On any successful connection, the teacher's assigned classes, rosters, active assessments and current term config are cached. Cache is tenant-namespaced (TEN-040) and cleared on logout.
Capture	Writes go to IndexedDB first, always. The UI never waits on the network to acknowledge a mark or a score.
Queue visibility	SyncLedger (§8.2) states the count and the state in words at all times on offline-capable screens.
Sync	Background sync on reconnection; manual retry available. Idempotency keys prevent double-posting on retry (NFR-API-010).
Conflict — same user, two devices	Last write by device timestamp wins, silently. Stated in the queue detail so it is discoverable.
Conflict — different users	Surfaced for manual reconciliation, never auto-resolved (FR-ATT-011). Both values, both authors, both times, one decision.

Concern	Behaviour
Failure	Named and actionable: what failed, why, what to do. Never a generic retry toast.
Honesty rule	No optimistic UI that implies server persistence. 'Saved on this phone' and 'saved to school' are different sentences and the interface always uses the true one.

9.2 PWA Infrastructure

FR-UX-010 requires the application to be installable as a PWA. v1.0 asserted this in the surfaces table and never specified a manifest, an icon set, or a caching strategy — and the companion prototype shipped as a plain static page with neither. Both were corrected in v1.1: the prototype ships with a real manifest.json and service worker (sw.js), described below rather than only claimed.

Element	Specification
Web app manifest	name 'PBSMS', short_name 'PBSMS', display 'standalone', start_url scoped to the Teacher Field App entry route, theme_color --pb-teal-900, background_color --pb-canvas, icons at 192px and 512px (maskable + any purpose).
Install prompt	Deferred beforeinstallprompt captured and surfaced as an in-app 'Install PBSMS' action after a teacher's second successful register submission, not on first load — install offers are more credible once the app has proven useful once.
Service worker — app shell	Cache-first for the shell, JS/CSS bundles and icon set, versioned and invalidated on deploy (a stale shell must never silently outlive a release).
Service worker — roster/assessment data	Network-first with a cache fallback, refreshed on every successful connection per §9.1's pre-fetch rule — never served stale-only.
Service worker — queued writes	Background Sync API registers pending attendance/score submissions; on 'sync' event, the queue in IndexedDB is flushed through the same idempotent endpoints §9.1 already specifies. The service worker is a delivery mechanism for the queue, not a second queue.

The PWA install icons at 192px and 512px are launcher artwork and are the one exception to §4.5's currentColor rule — a launcher icon is a raster asset generated at build time from the PBSMS mark, not a UI glyph, and is recorded in ICONS.md as such.

Honest limitation carried into this revision: a service worker requires an https or localhost origin to register — it will not activate if the prototype HTML is opened directly as a local file (file:///). The companion manifest.json and sw.js are real, are referenced from the prototype, and will activate when served through any local dev server (e.g. npx serve) or once deployed — but double-clicking the HTML file will not demonstrate installability. This is stated here rather than left for someone to discover by a failed install prompt.

9.3 Data-Saver Mode

FR-UX-040 requires 'a data-saver mode for report viewing.' v1.0 listed pagination and lazy image loading as general low-data practice but never specified this mode as its own concrete behaviour, on Parent View specifically, where it matters most.

- A visible toggle on any report-length screen (report card, statement of account): 'Data saver' — off by default on Wi-Fi-detected connections (via the Network Information API where available), on by default when the connection is detected as 2G/3G or save-data is signalled by the browser.
- With data saver on: photos (student photo, school logo) are replaced by initials/text placeholders; PDF attachments are not auto-fetched, replaced by a named download link with file size shown; charts on analytics screens render as a compact numeric table instead of an SVG chart.

- Icons are not affected by data saver. Inline SVG glyphs at the sizes in §4.5.2 are a few hundred bytes each and already in the shell bundle, and they are load-bearing for meaning under §4.5.4 — removing them would strip a WCAG 1.4.1 pairing to save nothing.
- Data saver never hides data, only defers heavier renderings of the same data — a parent on data saver still sees every score and every cedi, just not the school photo alongside it.
- The toggle's state persists per device, not per session, so a guardian on a fixed, expensive data plan does not re-decide this on every visit.

10. Role-Based Views

Chapter 13 defines scope from tenant down to record ownership. The interface renders permission-generated navigation (Chapter 34.1) — but per Chapter 34.5, hiding a control is a usability courtesy and never an authorization mechanism. The server decides; the UI merely declutters.

Role	Lands on	Sees	Does not see
Proprietor / Director	Group roll-up if multi-school, else school dashboard	All schools in tenant, finance summary, academic summary	Other tenants. Individual health and discipline detail unless separately granted.
Headmaster	Approvals & exceptions dashboard	Whole school: academics, attendance, finance summary, approvals	Platform settings, other schools unless assigned.
Accountant / Bursar	Finance dashboard	Invoices, payments, allocations, reconciliation, statements	Academic scores, health, discipline.
Academic Coordinator	Results workflow queue	Assessment config, results pipeline, curriculum, analytics	Finance, payroll, health.
Class / Subject Teacher	Teacher Field App home	Own assigned classes and subjects only	Other teachers' classes, finance, admin settings.
Admissions Officer	Application queue	Applications, duplicates, conversion	Enrolled-student finance and academic records beyond conversion.
Parent / Guardian	Parent home	Linked children only; published results, attendance, invoices, messages	Any other child. Unpublished or draft results.
Student	Student home	Own published results and timetable	Finance, discipline, other students.
Platform Support	Tenant list	One tenant at a time, under an active audited grant	Financial reversal, export, permission change without a second approver.

11. Accessibility Implementation

Chapter 42 binds the product to WCAG 2.1 Level AA with a CI gate (NFR-ACC-020) and a pre-GA manual audit (NFR-ACC-030). Naming a standard is not implementing one — this section is what conformance actually requires here.

11.1 Concrete commitments

WCAG criterion	Implementation in PBSMS
1.4.3 Contrast (Minimum)	All token pairs verified at 4.5:1 body / 3:1 large. Tenant brand colours validated on upload and rejected if they fail (§4.4).

WCAG criterion	Implementation in PBSMS
1.4.1 Use of Colour	Every status pairs colour with a glyph and a text label, per the register in §4.5.4. Attendance states, sync states, result states and payment states all readable in greyscale.
1.1.1 Non-text Content	Decorative icons beside a text label carry aria-hidden; an icon that is the only content of a control carries an accessible name matching its §4.5.4 meaning. Icon-only controls exist in exactly one place (§6.1.1's icon rail) and keep a tooltip as well.
2.1.1 Keyboard	Score grid, approval flows, allocation and every dialog fully operable by keyboard. The score grid is designed keyboard-first, not keyboard-compatible.
2.4.7 Focus Visible	2px focus ring in --pb-teal-600 with 2px offset, never removed. Focus is trapped in modals and returned on close.
2.5.5 Target Size	44x44 CSS px minimum on Teacher Field App and Parent View — the surfaces used on phones one-handed. An icon at --pb-icon-lg inside a 44px target, never a 24px tap area.
3.3.1 / 3.3.3 Errors	Errors name the field, state what is wrong and state how to fix it. Form-level summary lists every error with in-page links.
4.1.3 Status Messages	SyncLedger, toasts and validation summaries announce via aria-live so a screen-reader user learns a register synced without moving focus.
1.3.1 Info & Relationships	Data tables use real th/scope. The score grid uses a grid role with correct row and column semantics, not a div soup.

11.2 Verification, not assertion

Every contrast value in §4.1 must be re-checked against a contrast tool before implementation, and the CI accessibility job must fail the build on new critical violations from the first frontend commit — not added once the component library already exists. Retrofitting accessibility across an existing component set is the same category of expensive mistake as retrofitting multi-tenancy across an existing schema.

Fixed v1.0 gap: NFR-ACC-020 was written into this document's first revision but did not exist in the code scaffold's actual CI pipeline — a requirement stated in the spec and absent from the artifact meant to enforce it. A pa11y-ci job now exists in .github/workflows/ci.yml, running against the built Staff Console and Parent View, blocking merge on new critical violations. It is a starting gate, not the NFR-ACC-030 manual audit — the two remain distinct and both remain required before General Availability.

The icons:audit job added in §4.5.6 sits beside it and is the same kind of instrument: a rule that would otherwise decay into a convention, made into a build failure instead.

12. Performance Budget

NFR-PERF-020 requires p95 page load $\leq 2.5s$ on a 3G-equivalent connection. A budget makes that testable per screen rather than aspirational.

Surface	JS (gzipped)	CSS	Fonts	Distinct glyphs	Total first load	Rationale
Parent View	≤ 60 KB	≤ 15 KB	0 KB (system stack only)	≤ 12	≤ 150 KB	Metered data, arrives from a link, must never feel expensive to open.
Teacher Field App	≤ 140 KB	≤ 20 KB	≤ 28 KB	≤ 24	≤ 250 KB	Installed once, cached thereafter; offline shell justifies more.
Staff Console	≤ 250 KB	≤ 30 KB	≤ 28 KB	≤ 90	≤ 400 KB	Desktop, office connection, route-split per module.

Surface	JS (gzipped)	CSS	Fonts	Distinct glyphs	Total first load	Rationale
Platform Console	≤ 250 KB	≤ 30 KB	≤ 28 KB	≤ 40	≤ 400 KB	Internal use only.

- Icons count inside the JS figure, not beside it. Tree-shaken inline SVG at the sizes in §4.5.2 costs roughly 300–500 bytes gzipped per glyph, so the caps above are approximately 6KB on Parent View and 12KB on the Teacher Field App. An icon set that nobody counts is how a 150KB budget quietly becomes 190KB.
- The distinct-glyph column is a ceiling, not a target, and is enforced by the same CI budget check as the byte figures. Exceeding it is usually a sign that a screen is inventing meanings rather than reusing the register in §4.5.4.
- Route-level code splitting per module; no module's code loads for a user whose role cannot reach it.
- Lists paginate by default (FR-UX-040); no unbounded fetch anywhere.
- Images: student photos lazy-loaded, served at display size, never full-resolution originals in a list.
- Budget enforced in CI as a failing check, alongside the accessibility and icon-provenance gates.

13. Build Order

Sequenced so that the things that cannot be retrofitted — the offline layer, the accessibility baseline, and the token layer everything inherits from — exist before the component library grows.

Stage	Build	Why here
1	Tokens, type scale, icon set and icon tokens (§4.5), base components, focus and contrast baseline, CI accessibility and icon-provenance gates	Everything else inherits from this. Cheapest moment to get it right, and the last moment at which the icon set can be chosen once rather than argued about per component.
2	App shell, auth, context switcher, permission-generated navigation	Every screen mounts inside it.
3	Offline/sync layer + SyncLedger, proven with the register	Hardest to retrofit. Build it against one real screen before there are twenty.
4	Teacher Field App — register, score entry	Highest-frequency users; earliest real feedback; validates stage 3.
5	Staff Console — students, academic structure, assessment, results	The operational core.
6	Parent View — home, results, invoices, pay	Unlocks the communication value proposition.
7	Finance console — invoicing, payments, allocation, reconciliation	Depends on 5; highest correctness risk, so not first.
8	Communication, analytics, supporting operations, compliance screens	Breadth after depth.
9	Platform Console	Internal users tolerate rough edges longest; also gated on Chapter 3 impersonation controls existing.

The four domain glyphs of §4.5.5 are the only bespoke drawing on the critical path, and they belong in stage 1 with everything else in §4 — a placeholder glyph that ships into stage 5 becomes permanent.

14. Documented Deviations from the SRS

v1.0 contained one place where the actual design departed from the SRS's literal stated pattern while being presented as compliant. That is a worse failure than a missing feature — a missing feature is visible; a silent deviation is not. This section exists so it cannot happen again unnoticed: any future departure from the SRS's stated text belongs here, named, with the SRS clause, the deviation, and the reasoning, reviewed by whoever holds the Engineering Lead role before it ships.

SRS clause	Stated pattern	This design	Reasoning
34.5 — 'The frontend hides unavailable actions for usability'	Actions the current user cannot perform are removed from view	Impersonation-blocked actions in the Platform Console (§8.10) are shown, visibly disabled, with the four-eyes requirement named	34.5's own purpose is to prevent the client from being mistaken for the authority — the server still decides. For this specific screen, a support engineer who cannot see why a button is missing will assume the product is broken and open a second support ticket about the first one. Naming the constraint is more informative than hiding it, and does not weaken the actual authorization boundary, which remains server-enforced either way. This exception is scoped to the Platform Console only — every other surface follows 34.5's literal pattern of hiding, not disabling.

No new deviations arise from v1.2. The icon system in §4.5 satisfies Chapter 34.2's existing token requirement and introduces no departure from the SRS's stated text. This table is expected to grow as the product is actually built and real constraints surface — an empty deviations table after implementation would be a sign this section stopped being used, not a sign nothing was ever traded off.

15. Open Questions & Honest Limitations

This specification is a proposal by one designer working from a document, not from users. The following are genuinely unresolved and should not be treated as settled by the confidence of the prose above.

- No user research exists. Nothing here has been shown to a Ghanaian teacher, headmaster, bursar or parent. The register design in §8.1 is the highest-conviction screen and also the one most likely to be wrong in a way only a real teacher would catch. Test it before building stage 4.
- The four-surface split (§3) is the load-bearing structural bet. If it is wrong, most of this document changes. It should be reviewed by whoever holds the Engineering Lead role before stage 1 begins.
- Colour and type are proposals with reasoning, not brand decisions. If PBSMS acquires a brand identity later, §4.1 and §4.2 are the parts to replace — the rest of the document survives that change.
- Lucide (§4.5.1) is chosen on licence, grid and coverage, not on a comparison against Phosphor or Heroicons run on this product's actual screens. Any MIT set on a 24px grid with a variable stroke would satisfy the same constraints. If a review prefers another, §4.5.1 is a one-row change and nothing else in §4.5 moves — which is the point of specifying the rules rather than the icons.
- The semantic register (§4.5.4) is derived from the states this document already specifies, not from watching anyone use them. Whether teachers read 'excused' and 'sick' as two categories or one, and whether the distinction is worth two glyphs, belongs in the same stage-4 test as §8.1.
- The four domain glyphs (§4.5.5) do not exist yet. They are the only bespoke drawing this document requires and they sit on the stage-1 critical path.
- Contrast ratios in §4.1 are calculated, not tool-verified. Verify before implementation (§11.2). Icon strokes against --pb-canvas need the same check — a 1.5px stroke is a non-text contrast case under 1.4.11, and thin strokes fail more easily than filled text.

- The performance budgets in §12 are set from typical figures for this class of application, not from measurement of a PBSMS build that does not yet exist. The per-glyph byte estimate is likewise an estimate. Re-baseline once stage 4 ships, in the same spirit as NFR-PERF-090.
- Timetable generation is specified as conflict detection only, consistent with SRS Chapter 17 — automatic generation remains unspecified there and is deliberately not invented here.
- This document has been audited twice: once against the SRS's literal text (v1.0 → v1.1) and once against its own internal completeness (v1.1 → v1.2). Both audits were performed by the original author. A third audit by someone else would very likely find a different set of gaps again — the same blind spots that produce a gap tend to also miss it on self-review. Treat v1.2 as more accurate than v1.1, not as finished.

— *End of Document* —